

PROJECT REPORT

on

Online Voting System

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Declaration

I hereby declare that the project titled **“Online Voting System”** is an original work carried out by me as part of my academic requirements for the subject *Technical Training*.

This project has been developed by me under the guidance of Mrs. Gagandeep Kaur, and it has not been submitted previously to any other institution or university for the award of any degree or diploma.

I further declare that all sources of information used in this project have been properly acknowledged.

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1. Acknowledgement

I would like to express my sincere gratitude to my teacher, Mrs. Gagandeep Kaur for providing me with the opportunity to work on this project titled “Online Voting System”. Their guidance, encouragement, and support helped me successfully complete this project.

I would also like to thank my institution for providing the necessary resources and environment to carry out this work. Finally, I extend my thanks to my friends and family for their constant support and motivation.

2. Abstract

The Online Voting System is a console-based application developed using C++ that allows efficient management of voters and candidates while ensuring a fair voting process. The system enables users to add, update, delete, and view records of voters and candidates.

The application ensures that each voter can vote only once by maintaining a voting status. It uses file handling techniques to store and retrieve data, ensuring persistence even after the program is closed.

This project demonstrates the use of Object-Oriented Programming concepts along with fundamental data structures and file management techniques to build a real-world application.

3. Introduction

Voting is an essential part of any democratic system. Traditional voting methods often involve manual processes that can be time-consuming and prone to errors. The Online Voting System aims to simplify this process by providing a digital solution.

This project is designed as a simple console-based system where users can manage voters and candidates and conduct voting efficiently. The system ensures transparency, prevents duplicate voting, and maintains records using files.

The project is developed using C++ and incorporates concepts such as classes, arrays, file handling, and basic algorithms.

4. System Design/Methodology

System Overview

The system is divided into two main components:

- Candidate Management
- Voter Management

Modules

1. Add Candidate
2. Add Voter
3. Voting Process
4. Show Results
5. Show Winner
6. Update/Delete Records

Data Structures Used

- Arrays for storing voters and candidates
- Linear search for finding records
- File handling for persistent storage

Algorithm Flow

1. Load data from files at startup
2. Display menu
3. Perform selected operation
4. Save updated data to files
5. Repeat until exit

5.Implementation

The project is implemented using C++ with the following key features:

Classes Used

- **Candidate Class**
 - ID
 - Name
 - Votes
- **Voter Class**
 - ID
 - Name
 - Voting status (hasVoted)

Core Functionalities

- Add new voters and candidates
- Modify and delete records
- Voting system ensuring one vote per user
- Display results and winner

File Handling

- Data is stored in:
 - candidates.txt
 - voters.txt
- Files are updated after every operation

Programming Concepts Used

- Object-Oriented Programming (OOP)
- Arrays
- File handling (ifstream, ofstream)
- Conditional statements and loops

Code

```
#include <iostream>
#include <fstream>
using namespace std;

class Candidate {
public:
    int id;
    string name;
    int votes;

    void input() {
        cout << "Enter Candidate ID and Name: ";
        cin >> id >> name;
        votes = 0;
    }
}
```

```

}
void display() {
    cout << id << " - " << name << " | Votes: " << votes << endl;
}
};

class Voter {
public:
    int id;
    string name;
    bool hasVoted;

    void input() {
        cout << "Enter Voter ID and Name: ";
        cin >> id >> name;
        hasVoted = false;
    }

    void display() {
        cout << id << " - " << name << " | Voted: " << (hasVoted ? "Yes" : "No") << endl;
    }
};

Candidate candidates[100];
Voter voters[100];
int numCandidates = 0, numVoters = 0;

// ----- FILE HANDLING -----
void saveCandidates() {
    ofstream file("candidates.txt");
    for (int i = 0; i < numCandidates; i++) {
        file << candidates[i].id << " " << candidates[i].name << " " << candidates[i].votes << endl;
    }
}

void loadCandidates() {
    ifstream file("candidates.txt");
    while (file >> candidates[numCandidates].id >> candidates[numCandidates].name >>
candidates[numCandidates].votes) {
        numCandidates++;
    }
}

void saveVoters() {
    ofstream file("voters.txt");
    for (int i = 0; i < numVoters; i++) {
        file << voters[i].id << " " << voters[i].name << " " << voters[i].hasVoted << endl;
    }
}

void loadVoters() {
    ifstream file("voters.txt");
    while (file >> voters[numVoters].id >> voters[numVoters].name >> voters[numVoters].hasVoted) {
        numVoters++;
    }
}

// ----- SHOW -----
void showCandidates() {

```

```

    cout << "\n--- Candidates ---\n";
    for (int i = 0; i < numCandidates; i++) {
        candidates[i].display();
    }
}

void showVoters() {
    cout << "\n--- Voters ---\n";
    for (int i = 0; i < numVoters; i++) {
        voters[i].display();
    }
}

// ----- DELETE -----
void deleteCandidate(int id) {
    for (int i = 0; i < numCandidates; i++) {
        if (candidates[i].id == id) {
            for (int j = i; j < numCandidates - 1; j++) {
                candidates[j] = candidates[j + 1];
            }
            numCandidates--;
            saveCandidates();
            cout << "Candidate deleted!\n";
            return;
        }
    }
    cout << "Candidate not found!\n";
}

void deleteVoter(int id) {
    for (int i = 0; i < numVoters; i++) {
        if (voters[i].id == id) {
            for (int j = i; j < numVoters - 1; j++) {
                voters[j] = voters[j + 1];
            }
            numVoters--;
            saveVoters();
            cout << "Voter deleted!\n";
            return;
        }
    }
    cout << "Voter not found!\n";
}

// ----- UPDATE -----
void updateCandidate(int id) {
    for (int i = 0; i < numCandidates; i++) {
        if (candidates[i].id == id) {
            cout << "Enter new name: ";
            cin >> candidates[i].name;
            saveCandidates();
            cout << "Updated!\n";
            return;
        }
    }
    cout << "Candidate not found!\n";
}

```

```

void updateVoter(int id) {
    for (int i = 0; i < numVoters; i++) {
        if (voters[i].id == id) {
            cout << "Enter new name: ";
            cin >> voters[i].name;
            saveVoters();
            cout << "Updated!\n";
            return;
        }
    }
    cout << "Voter not found!\n";
}

// ----- MAIN -----
int main() {

    loadCandidates();
    loadVoters();

    int choice;

    do {
        cout << "\n===== ONLINE VOTING SYSTEM =====\n";
        cout << "1. Add Candidate\n";
        cout << "2. Add Voter\n";
        cout << "3. Vote\n";
        cout << "4. Show Results\n";
        cout << "5. Show Candidates\n";
        cout << "6. Show Voters\n";
        cout << "7. Delete Candidate\n";
        cout << "8. Delete Voter\n";
        cout << "9. Update Candidate\n";
        cout << "10. Update Voter\n";
        cout << "11. Show Winner\n";
        cout << "12. Exit\n";
        cout << "Enter choice: ";
        cin >> choice;

        if (choice == 1) {
            candidates[numCandidates].input();
            numCandidates++;
            saveCandidates();
        }

        else if (choice == 2) {
            voters[numVoters].input();
            numVoters++;
            saveVoters();
        }

        else if (choice == 3) {
            int vid;
            cout << "Enter Voter ID: ";
            cin >> vid;

            for (int i = 0; i < numVoters; i++) {
                if (voters[i].id == vid) {

```

6.Output

The program provides a menu-driven interface with options such as:

- Add Candidate
- Add Voter
- Vote
- Show Results
- Show Candidates
- Show Voters
- Update/Delete Records
- Show Winner

```
===== ONLINE VOTING SYSTEM =====
```

1. Add Candidate
2. Add Voter
3. Vote
4. Show Results
5. Show Candidates
6. Show Voters
7. Delete Candidate
8. Delete Voter
9. Update Candidate
10. Update Voter
11. Show Winner
12. Exit

```
Enter choice: 5
```

```
PS C:\Users\salon\OneDrive\Desktop\TT-Project> ./voting
```

1. Add Candidate
2. Add Voter
3. Vote
4. Show Results
5. Show Candidates
6. Show Voters
7. Delete Candidate
8. Delete Voter
9. Update Candidate
10. Update Voter
11. Show Winner
12. Exit

```
Enter choice: 5
```

```
===== ONLINE VOTING SYSTEM =====
```

1. Add Candidate
2. Add Voter
3. Vote
4. Show Results
5. Show Candidates
6. Show Voters
7. Delete Candidate
8. Delete Voter
9. Update Candidate
10. Update Voter
11. Show Winner
12. Exit

```
Enter choice: 11
```

```
Winner: Ravi (5 votes)
```

3. Vote
4. Show Results
5. Show Candidates
6. Show Voters
7. Delete Candidate
8. Delete Voter
9. Update Candidate
10. Update Voter
11. Show Winner
12. Exit

```
Enter choice: 6
```

```
--- Voters ---
```

- | | |
|-------|------------|
| 1 - a | Voted: Yes |
| 2 - b | Voted: Yes |
| 3 - c | Voted: Yes |
| 4 - d | Voted: Yes |
| 5 - e | Voted: Yes |
| 6 - f | Voted: Yes |

7. RESULTS AND DISCUSSION

The system successfully performs all required operations including adding, updating, deleting, and displaying records. It ensures that each voter can vote only once by maintaining a voting flag.

The use of file handling ensures that data is not lost after program termination. The system is efficient for small-scale applications but uses linear search, which has a time complexity of $O(n)$.

The project demonstrates a clear understanding of programming fundamentals and problem-solving techniques.

8. CONCLUSION

The Online Voting System project successfully demonstrates the implementation of a real-world application using C++. It integrates Object-Oriented Programming, file handling, and data structures to provide a functional voting system.

Although the system is basic, it can be further enhanced with advanced data structures, graphical user interfaces, and database integration for larger-scale applications.

9. REFERENCES

Github Link: <https://github.com/Saloni-Singh01/Online-Voting.git>

- C++ Programming Language – Bjarne Stroustrup
- Object-Oriented Programming Concepts
- Online tutorials and documentation
- Visual Studio Code documentation
- C++ File Handling (ifstream, ofstream) resources